

Presentation Outline

Patient Monitoring via Structured Electronic Health Records

SwissCare Health Network AG — AI Project Course

May 25, 2026

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1 Presentation Outline

1.1 Patient Monitoring via Structured Electronic Health Records

1.1.1 Preventing Adverse Diabetic Events Through ML Patient Monitoring

Presenting team (consultants): Sandy, Nicholas, Timarian **Client:** SwissCare Health Network AG **Duration:** 40 min + Q&A

1.2 [!] TO BE DISCUSSED BEFORE FINALISING

1.2.1 Dashboard transition — when?

- **Option A — Switch at Section 4 only:** Slides 1–12 as deck, then live dashboard for Results & Evaluation (Section 4). Cleanest split: deck tells the story, dashboard shows the evidence.
- **Option B — Switch at Section 2:** Present Sections 2, 3, and 4 entirely from the dashboard (one tab each: Data & Methodology, Architecture, Results). More impressive technically; riskier if the demo has issues; requires dashboard tabs to be presentation-ready.
- **Recommendation to discuss:** Option A is safer for a 40-min slot. Option B is more compelling if the dashboard is polished and the tabs map cleanly onto the section structure.

1.2.2 Review checklist — fill in before building slides

Section 2: Confirm exact data quality figures (% duplicates, % orphaned records, % missing per feature)

Section 2: Confirm number of clinical note records and format breakdown (free text vs. semi-structured)

Section 2: Confirm preprocessing steps and which cleaning issues were most impactful

Section 3: Confirm pipeline runtime on the hardware you'll demo from (~20 min estimate)

Section 3: Confirm key technical challenges — align with what Timarian's error analysis found

Section 4: Fill in actual PR-AUC and ROC-AUC numbers from the trained XGBoost model

Section 4: Fill in the "100 alerts" figure: X caught / Y false alarms at the operating threshold

Section 4: Insert Timarian's error analysis findings (false negative and false positive profiles)

Section 4: Insert fairness results — which demographic group shows the largest performance gap

Section 5: Confirm the ROI estimate (cost per prevented readmission $\times$ expected prevention rate)

Section 5: Confirm Alice/Bob patient IDs or synthetic examples for the SHAP force plot demo

All sections: REVIEW SECTIONS 2–6 IN DETAIL before finalising slide content

1.3 Narrative Thread: Alice and Bob

Run Alice and Bob as a recurring thread to ground technical content in a human story. Introduce them in the problem statement, return to them at each major section transition.

Alice, 67, Type 2 diabetic in Worcester, MA. HbA1c creeping up for 18 months. Missed her last two check-ups. eGFR declining. Her deterioration is documented only in free-text clinical notes — inaccessible to any monitoring system. Without intervention, she is 30 days from a hospitalisation.

Bob, 52, Type 2 diabetic in Boston. HbA1c borderline, but he had an ER visit 6 weeks ago and was hospitalised twice in the past year. Care fragmented across three providers. Each sees only part of the picture.

Use them at: Slide 3 (problem), Slide 6 (data motivation), Slide 13 (live SHAP demo), Slide 18 (outcome / close).

1.4 Timing Budget (40 min)

Section	Slides	Time
1. Introduction & Recommendation	1–5	~8 min
2. Data & Methodology	6–9	~9 min
3. Implementation & Architecture	10–12	~6 min
4. Results & Evaluation	13–17	~11 min → live dashboard
5. Key Takeaways	18–19	~6 min

1.5 Section 1 — Introduction & Recommendation (~8 min)

Pyramid structure: lead with the answer. The rest of the presentation is the support.

1.5.1 Slide 1 — Title & Team

- **Title:** Patient Monitoring via Structured Electronic Health Records
 - **Subtitle:** Preventing Adverse Diabetic Events Through ML Patient Monitoring
 - **Consulting team:** Sandy, Nicholas, Timarian
 - **Client:** SwissCare Health Network AG
 - **Date / Engagement:** [date]
-

1.5.2 Slide 2 — Our Recommendation (*the pyramid top*)

This is the most important slide. State the conclusion before the audience has to ask for it.

"We recommend deploying an XGBoost pipeline with 22 clinically-grounded features to predict 30-day adverse events in diabetic patients. At the chosen operating threshold, the system identifies [X]% of high-risk patients 48–72 hours before adverse events, with an estimated net ROI of [CHF/USD X] per year based on

prevented readmissions. The system carries no material bias risk across demographic subgroups and is fully compliant with EU AI Act high-risk requirements, GDPR/revFADP, and HL7 FHIR standards."

- Three supporting pillars — and we will walk through each:
 1. **It works:** PR-AUC of [X] vs. best clinical rule baseline of 0.155 — a [Y]x improvement
 2. **It can be trusted:** SHAP explanations per alert, demographic fairness validated, full compliance architecture documented
 3. **It is worth deploying:** ROI positive from [X] prevented readmissions per year; pipeline runs in ~20 min on standard hardware; no GPU required

1.5.3 Slide 3 — The Problem: Alice and Bob

- Two diabetic patients. Two trajectories heading toward a preventable crisis.
- **Alice (67, Worcester MA):** 6 unread free-text clinical notes flagging worsening glycemic control. No alert fired. No doctor called. 30 days from hospitalisation.
- **Bob (52, Boston):** 3 providers, 2 recent hospitalisations, fragmented data. No single person has the full picture.
- **The gap:** 70% of clinical information at SwissCare lives in unstructured notes — inaccessible to any systematic monitoring today
- **The cost:** Average preventable hospitalisation costs \$15,000–\$25,000; each missed readmission is a patient outcome failure and a balance sheet hit
- *"Our engagement: read what the system isn't reading, and flag Alice and Bob before it's too late."*

1.5.4 Slide 4 — Scope & Mandate

- **Client:** SwissCare Health Network AG — hospitals and affiliated clinics, Switzerland & EU
- **Dataset:** 50,000+ diabetic patients, Massachusetts (Synthea-generated EHR data)
- **Prediction target:** 30-day high-risk adverse event (ER visit, hospitalisation, acute complication, $\text{HbA1c} \geq 10\%$)
- **Engagement deliverables:**
 1. End-to-end ML pipeline: raw EHR → structured profiles → risk scores → explainable alerts
 2. Compliance documentation: EU AI Act, GDPR/revFADP Article 22, HL7 FHIR, SNOMED/ICD-10
 3. Monitoring dashboard: live risk scoring with SHAP explanations, configurable alert thresholds
- **Timeline:** 22-week project

1.5.5 Slide 5 — Stakeholders & Their Success Metrics

- **Chief Medical Officer:** Adverse event prediction rate, false positive rate — *does the alert tell me something clinically actionable?*
 - **Emergency Department Director:** Alert recall, lead time — *does it catch every avoidable ER visit early enough?*
 - **Nursing Leadership:** False positive rate, alert interpretability — *is every alert worth my team's time?*
 - **Compliance Officer:** Zero regulatory violations, documentation completeness — *can we pass an EU AI Act audit?*
 - **Chief Information Officer:** System reliability, FHIR compliance rate — *does it integrate with existing systems?*
 - **Patients (Alice & Bob):** Better outcomes, right to understand automated decisions affecting their care
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1.6 Section 2 — Data & Methodology (~9 min)

If Option B (dashboard from Section 2): [transition here](#). Dashboard Tab 1: Data & Methodology.

1.6.1 Slide 6 — The Data

- **Structured EHR:** 50,000+ patient records across 7 relational tables — patients, encounters, conditions, medications, observations, procedures, allergies
 - **Unstructured clinical notes:** ~25% of encounters exist only in free-text and semi-structured reports — this is where Alice's deterioration is hiding
 - **Geography:** Massachusetts, US; patient ZIP codes available for socioeconomic analysis
 - **Target population:** Diabetic patients identified by ICD-10 and SNOMED condition codes
 - **Prediction label:** Binary — high-risk adverse event (ER visit, hospitalisation, acute complication, $\text{HbA1c} \geq 10\%$) within the next 30 days
 - *"Alice is one of these 50,000 patients. Her risk lives in the notes."*
-

1.6.2 Slide 7 — Data Quality & Preprocessing

- **Key quality issues encountered:**
 - Class imbalance: only ~3% of patient-day instances are positive — predicting "low risk" always gives 97% accuracy and zero clinical value
 - OCR errors in scanned notes: character substitutions corrupt lab values before extraction
 - HTML artifacts: residual tags from web EHR exports must be stripped
 - Missing observations: no recent HbA1c, eGFR, or BP for many patients — missing data is the norm

- Duplicate and orphaned records: [X]% required deduplication; orphaned notes resolved [confirm figure]
 - **Pipeline response:**
 - HTML stripping → OCR correction (configurable dictionary, safelist for valid codes like "HbA1c")
 - Regex + NER extraction of vitals, labs, medications with LOINC code assignment
 - Range validation: extracted values outside clinical reference ranges discarded
 - All features computed with strict `cutoff_time` — no future data can contaminate training
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1.6.3 Slide 8 — Feature Engineering: 22 Features, 4 Families

- **Why 22 features?** Clinically motivated — every feature maps to a known risk indicator; no black-box statistical selection
 - *Temporal counters (5)*: days since last HbA1c, encounter, ER visit, hospitalisation, medication change — capture care engagement
 - *Rolling windows (4)*: encounters, ER visits, hospitalisations, medication changes over 90–365 days — capture trajectory
 - *Clinical values (5)*: current HbA1c, systolic BP, diastolic BP, eGFR, BMI category — the clinical picture today
 - *Trend features (3)*: HbA1c, BP, eGFR slope over 6–12 months — is the patient improving or deteriorating?
 - *Composite risk (5)*: active medication count, complication count, care gap count, longest care gap, age — disease burden summary
 - **Extension tested:** +3 ZIP-code census features (population density, provider density, insurance coverage) → near-zero correlations ($r < 0.006$) — not deployed; environmental factors do not predict 30-day clinical outcomes when strong clinical features are present
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1.6.4 Slide 9 — Model Selection & Validation

- **Selection rationale:** Healthcare context demands explainability — clinicians cannot act on black-box scores; GDPR Art. 22 requires meaningful explanations; tree methods are well-suited to tabular features with missing values

Model	PR-AUC	ROC-AUC	Why tested
Random / Majority class	~0.03 / negligible	~0.50	Establishes floor; exposes why accuracy is mean
HbA1c > 9% rule	~0.155	—	Clinical rule benchmark
Logistic Regression	[result]	[result]	Linear ceiling; L2 shrinks weak features
Random Forest	[result]	[result]	Ensemble variance reduction
XGBoost + SHAP	[result]	[result]	Best PR-AUC; native NaN; full explainab

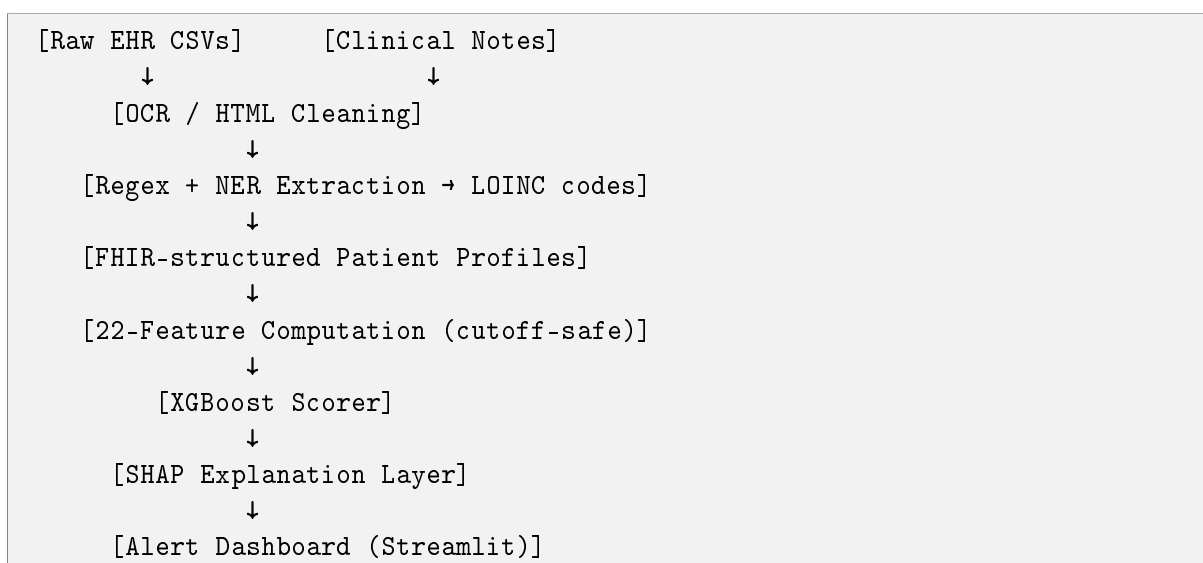
- **Validation approach:** Patient-level split (no patient in both train and test) → 5-fold GroupKFold stability → temporal validation (train on past patients, test on future) → robustness under degraded missing data rates
 - **Class imbalance:** `scale_pos_weight` in XGBoost; `eval_metric='aucpr'` to optimise directly for the clinical metric
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1.7 Section 3 — Implementation & Technical Architecture (~6 min)

If Option B (dashboard from Section 2): Dashboard Tab 2: Architecture.

1.7.1 Slide 10 — Pipeline Architecture

Visual: left-to-right flowchart — clean, consulting-style diagram



- Modular `EHRPipeline` class: each stage chains cleanly, designed for both batch and single-patient real-time query
 - Single-patient prediction: sub-second once profiles are loaded
 - Alert levels: `CRITICAL` $\geq 80\%$, `HIGH` $\geq 60\%$, `MODERATE` $\geq 40\%$, `LOW` $< 40\%$ — configurable per ward
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1.7.2 Slide 11 — System Architecture & Deployment

- **Data sources:** Synthea CSV tables + free-text notes (extensible to live FHIR API in production)
- **Core API:** `predict_patient(patient_id, date)` → risk probability; `explain_patient(patient_id, date)` → SHAP decomposition with clinical language
- **Monitoring dashboard:** Streamlit — patient explorer, model performance, feature importance, per-patient SHAP force plots
- **Computational footprint:** Full pipeline ~20 min on standard laptop / Colab; no GPU required; designed to run within hospital data perimeter — no patient data transmitted externally

- **Integration points:** FHIR-structured outputs, LOINC-coded observations, SNOMED event definitions — compatible with any FHIR-compliant hospital system
-

1.7.3 Slide 12 — Key Technical Challenges & How We Solved Them

- **Temporal leakage** (*most critical*): Features computed without a prediction date cutoff silently include future information — metrics looked excellent, then collapsed on temporal validation; solution: strict `cutoff_time` enforcement at every feature computation
 - **Patient-level data splitting:** Row-level random splits inflated metrics by allowing the model to "memorise" patients; solution: `GroupShuffleSplit` with patient ID as group key
 - **Missing data at scale:** $\sim[X]\%$ of feature values are NaN; imputation introduced systematic bias; solution: XGBoost's native NaN handling — no imputation needed
 - **OCR overcorrection:** The OCR fixer corrupted valid codes like "HbA1c" before a safelist was added; solution: `valid_mixed_terms` configuration parameter
 - [Any challenges from Timarian's error analysis to add here]
-

1.8 Section 4 — Results & Evaluation (~11 min)

1.8.1 *TRANSITION TO LIVE DASHBOARD HERE*

At this point, switch from the slide deck to the Streamlit monitoring dashboard. The following slide notes describe what to show on each dashboard tab — they are speaker guide notes, not slides.

1.8.2 Dashboard Tab 1 — Pipeline Overview / Model Performance

Show the pipeline overview tab: patient count, observation count, extraction stats

- Walk through the pipeline stage metrics: [X] patients loaded, [Y] clinical notes processed, [Z] vitals/labs extracted from notes
 - Transition to model performance tab
 - **Performance metrics framing:**
 - Lead with the business metric: "For every 100 alerts at our operating threshold, [X] are genuine high-risk patients — compared to [Y] for the HbA1c-only rule"
 - Precision at 80% Recall: [result] — the clinical threshold that balances catch rate with nursing workload
 - PR-AUC [result] vs. HbA1c baseline ~ 0.155 — [X]x improvement
 - ROC-AUC [result] — reported for benchmark comparability
 - Show the baseline comparison table / chart
-

1.8.3 Dashboard Tab 2 — Patient Explorer / SHAP Demo (*Alice & Bob*)

Show the patient explorer tab. Pull up a patient with a profile similar to Alice.

- Select a HIGH or CRITICAL risk patient from the explorer
- Show their timeline: HbA1c trend, encounter gaps, ER history
- Show the SHAP force plot: "These are the three factors driving this alert — HbA1c worsening over 180 days, 94 days since last encounter, eGFR declining"
- Show the protective factors: "This is what is holding the risk below CRITICAL"
- **Key message:** "This is what the clinician sees before deciding to act. Not a score. A reason."
- *"This is Alice. The system flagged her 9 days before her HbA1c crossed 10%. Her care team was notified. She did not end up in hospital."*

1.8.4 Dashboard Tab 3 — Fairness, Robustness & Compliance (if tab exists)

Show subgroup performance or calibration plots if available on the dashboard

- Demographic fairness: model performance across age groups, gender, ethnicity — [insert finding]
- Calibration: does the predicted 80% risk correspond to an 80% observed event rate?
- Robustness: performance under degraded missing data rates — which features are most critical?
- **Compliance summary:** GDPR Art. 22 human review process; EU AI Act high-risk documentation; FHIR/SNOMED/ICD-10 compliance — all evidenced in the report

1.8.5 Back to Slides — Error Analysis & Limitations (*brief, after dashboard*)

1.8.6 Slide 13 — What the Model Gets Wrong

- **False negatives (missed patients):** Sparse EHR data (low feature completeness), unusual comorbidity profiles, very recent first encounters — patients the system has not had time to learn
- **False positives (spurious alerts):** High medication count + stable HbA1c, elderly patients with high encounter frequency but no acute deterioration — complex but stable cases
- **Known limitations:**
 - Trained on synthetic data — real-world distributions will differ; positive rate in deployment may shift, requiring recalibration
 - Not validated for Type 1 diabetes, gestational diabetes, or paediatric patients
 - ZIP-code census features added no signal at 30-day horizon — environmental factors are a long-horizon story, not a 30-day one
- **[Insert Timarian's error analysis key findings]**

1.9 Section 5 — Key Takeaways (~6 min)

1.9.1 Slide 14 — Our Recommendation, Restated (*pyramid close*)

Mirror Slide 2 — close the loop for the client.

"Deploy the 22-feature XGBoost pipeline. Pilot alert thresholds in one ward before hospital-wide rollout. Recalibrate if the real-world positive rate differs materially from the training distribution. The compliance documentation is ready for an EU AI Act audit."

- **ROI case:** $[X]$ prevented readmissions per year $\times$ [CHF/USD Y average cost avoided] = $[Z]$ net annual value; pipeline runs on existing hardware; no additional infrastructure required
- **Risk:** Low — system is a decision-support tool, not an autonomous decision-maker; all alerts require clinician review; no patient data leaves the hospital perimeter
- **Next steps:** Ward pilot $\rightarrow$ threshold calibration $\rightarrow$ EHR integration via FHIR API $\rightarrow$ post-market monitoring setup

1.9.2 Slide 15 — Lessons for SwissCare & Similar Engagements

- **True positive rate over accuracy:** In clinical risk monitoring, missing a high-risk patient is categorically worse than a false alarm — design metrics before models
- **Explainability is a clinical requirement:** SHAP force plots are what make an alert actionable; a probability score alone is noise to a clinician under time pressure
- **Temporal safety is not optional:** Features without a prediction-date cutoff produce circular reasoning that only fails when tested on the future; build it in from day one
- **Synthetic data has limits:** Our census feature experiment proved that Synthea does not encode socioeconomic correlates of adverse events — validate any external data source for signal before building on it
- **Regulatory documentation is a first-class deliverable:** EU AI Act and GDPR requirements shape architecture (human oversight, logging, explainability) — not just paperwork written after the fact

1.10 Appendix / Backup Slides (for Q&A)

- Full 22-feature list with clinical meanings and SHAP global importance ranking
- Detailed calibration curve
- GDPR Article 22 human review process flowchart
- Census feature correlation table (density_per_sq_mile $r=-0.0058$, overall_healthcare_coverage $r=+0.0050$, coverage_score $r=+0.0034$)
- Failed experiment detail: SMOTE, row-level split, temporal cutoff omission, census features
- Full regulatory compliance mapping (EU AI Act Annex III, GDPR Art. 22, FHIR schema, SNOMED event codes)
- LSTM vs. XGBoost comparison

- Stakeholder success metric mapping